Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.20)

Metric

$$ds^2 = \left(-ze^{\left(2\,\widetilde{U}\right)} - e^{\left(2\,\widetilde{U}\right)}r\right)\mathrm{d}t \otimes \mathrm{d}t + \frac{e^{(2\,\lambda)}}{2\,r}\mathrm{d}\rho \otimes \mathrm{d}\rho + \left(\frac{\rho^2}{ze^{\left(2\,\widetilde{U}\right)} + e^{\left(2\,\widetilde{U}\right)}r}\right)\mathrm{d}\varphi \otimes \mathrm{d}\varphi + \frac{e^{(2\,\lambda)}}{2\,r}\mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, \rho, \varphi, z)$$

Metric Functions

$$U = -\frac{m_1}{r_1} - \frac{m_2}{r_2}$$

$$r = b_A^2 + \rho^2 - 2\,b_A z + z^2$$

$$\lambda = \frac{2\,b_1 b_2 m_1 m_2}{(b_1^2 - 2\,b_1 b_2 + b_2^2)r_1 r_2} + \frac{2\,m_1 m_2 \rho^2}{(b_1^2 - 2\,b_1 b_2 + b_2^2)r_1 r_2} - \frac{2\,b_1 m_1 m_2 z}{(b_1^2 - 2\,b_1 b_2 + b_2^2)r_1 r_2} - \frac{2\,b_2 m_1 m_2 z}{(b_1^2 - 2\,b_1 b_2 + b_2^2)r_1 r_2} + \frac{2\,m_1 m_2 z^2}{(b_1^2 - 2\,b_1 b_2 + b_2^2)r_1 r_2} - \frac{m_1^2 \rho^2}{2\,r_1^4} - \frac{m_2^2 \rho^2}{2\,r_2^4}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\,\sqrt{2}\sqrt{z + r}e^{\left(\widetilde{U}\right)}\right)\,dt + \left(\frac{e^{(\lambda)}}{2\,\sqrt{r}}\right)\,d\rho$$

$$n = \left(\frac{1}{2}\,\sqrt{2}\sqrt{z + r}e^{\left(\widetilde{U}\right)}\right)\,dt + \left(-\frac{e^{(\lambda)}}{2\,\sqrt{r}}\right)\,d\rho$$

$$m = \left(\frac{\sqrt{2}\rho e^{\left(-\widetilde{U}\right)}}{2\sqrt{z+r}} \right) d\varphi + \left(\frac{i\, e^{(\lambda)}}{2\sqrt{r}} \right) dz$$

$$\overline{m} = \left(\frac{\sqrt{2}\rho e^{\left(-\widetilde{U}\right)}}{2\sqrt{z+r}} \right) d\varphi + \left(-\frac{i\, e^{(\lambda)}}{2\sqrt{r}} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{z+r}}{\sqrt{2}ze^{\left(\widetilde{U}\right)} + \sqrt{2}e^{\left(\widetilde{U}\right)}{}_r} \right) dt + \left(e^{(-\lambda)}\sqrt{r} \right) d\rho$$

$$n = \left(-\frac{\sqrt{z+r}}{\sqrt{2}ze^{\left(\widetilde{U}\right)} + \sqrt{2}e^{\left(\widetilde{U}\right)}{}_r} \right) dt + \left(-e^{(-\lambda)}\sqrt{r} \right) d\rho$$

$$m = \left(\frac{\sqrt{2}ze^{\left(\widetilde{U}\right)}}{2\,\rho\sqrt{z+r}} + \frac{\sqrt{2}e^{\left(\widetilde{U}\right)}{}_r}{2\,\rho\sqrt{z+r}} \right) d\varphi + \left(i\, e^{(-\lambda)}\sqrt{r} \right) dz$$

$$\overline{m} = \left(\frac{\sqrt{2}ze^{\left(\widetilde{U}\right)}}{2\,\rho\sqrt{z+r}} + \frac{\sqrt{2}e^{\left(\widetilde{U}\right)}{}_r}{2\,\rho\sqrt{z+r}} \right) d\varphi + \left(-i\, e^{(-\lambda)}\sqrt{r} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{z+r}}{\sqrt{2}ze^{\left(\widetilde{U}\right)} + \sqrt{2}e^{\left(\widetilde{U}\right)}{}_r}\, \partial_t + e^{(-\lambda)}\sqrt{r}\, \partial_\rho$$

$$\Delta = -\frac{\sqrt{z+r}}{\sqrt{2}ze^{\left(\widetilde{U}\right)} + \sqrt{2}e^{\left(\widetilde{U}\right)}{}_r}\, \partial_t - e^{(-\lambda)}\sqrt{r}\, \partial_\rho$$

$$\delta = \frac{\sqrt{2} z e^{\left(\widetilde{U}\right)}}{2\,\rho\sqrt{z+r}} + \frac{\sqrt{2} e^{\left(\widetilde{U}\right)} r}{2\,\rho\sqrt{z+r}}\,\partial_{\varphi} + i\,e^{(-\lambda)}\sqrt{r}\,\partial_{z}$$

$$\overline{\delta} = \frac{\sqrt{2} z e^{\left(\widetilde{U}\right)}}{2\,\rho\sqrt{z+r}} + \frac{\sqrt{2} e^{\left(\widetilde{U}\right)} r}{2\,\rho\sqrt{z+r}}\,\partial_{\varphi} - i\,e^{(-\lambda)}\sqrt{r}\,\partial_{z}$$

Spin Coefficients

$$\rho = \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$
$$+ \frac{\rho z\frac{\partial}{\partial\rho} r}{4\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)\sqrt{r}} + \frac{\rho\sqrt{r}\frac{\partial}{\partial\rho} r}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{z\sqrt{r}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{r^{\frac{3}{2}}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$

$$\sigma = \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho z\frac{\partial}{\partial\rho} r}{4\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)\sqrt{r}} - \frac{z\sqrt{r}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{r^{\frac{3}{2}}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$

$$\kappa = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} + \frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\lambda}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} + \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\lambda}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\,z\frac{\partial}{\partial z} r}{4\left(z e^{(\lambda)} + e^{(\lambda)} r\right)\sqrt{r}} - \frac{i\sqrt{r}\frac{\partial}{\partial z} r}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\sqrt{r}}{4\left(z e^{(\lambda)} + e^{(\lambda)} r\right)}$$

$$\mu = \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$
$$+ \frac{\rho z\frac{\partial}{\partial\rho} r}{4\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)\sqrt{r}} + \frac{\rho\sqrt{r}\frac{\partial}{\partial\rho} r}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{z\sqrt{r}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{r^{\frac{3}{2}}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$

$$\lambda = \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho z\sqrt{r}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} + \frac{\rho r^{\frac{3}{2}}\frac{\partial}{\partial\rho}\lambda}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{\rho z\frac{\partial}{\partial\rho} r}{4\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)\sqrt{r}} - \frac{z\sqrt{r}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)} - \frac{r^{\frac{3}{2}}}{2\left(\rho z e^{(\lambda)} + \rho e^{(\lambda)} r\right)}$$

$$\nu = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} + \frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\lambda}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} + \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\lambda}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\,z\frac{\partial}{\partial z} r}{4\left(z e^{(\lambda)} + e^{(\lambda)} r\right)\sqrt{r}} - \frac{i\sqrt{r}\frac{\partial}{\partial z} r}{2\left(z e^{(\lambda)} + e^{(\lambda)} r\right)} - \frac{i\sqrt{r}}{4\left(z e^{(\lambda)} + e^{(\lambda)} r\right)}$$

$$\alpha = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,\sqrt{r}\frac{\partial}{\partial z}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,\sqrt{r}}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

$$\beta = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,\sqrt{r}\frac{\partial}{\partial z}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,\sqrt{r}}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

$$\tau = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\lambda}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\lambda}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{i\,z\frac{\partial}{\partial z}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)\sqrt{r}} - \frac{i\,\sqrt{r}}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

$$\pi = -\frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,z\sqrt{r}\frac{\partial}{\partial z}\lambda}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} - \frac{i\,r^{\frac{3}{2}}\frac{\partial}{\partial z}\lambda}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{i\,z\frac{\partial}{\partial z}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)\sqrt{r}} - \frac{i\,\sqrt{r}}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

$$\epsilon = \frac{z\sqrt{r}\frac{\partial}{\partial \rho}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{r^{\frac{3}{2}}\frac{\partial}{\partial \rho}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{\sqrt{r}\frac{\partial}{\partial \rho}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

$$\gamma = \frac{z\sqrt{r}\frac{\partial}{\partial \rho}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{r^{\frac{3}{2}}\frac{\partial}{\partial \rho}\widetilde{U}}{2\left(ze^{(\lambda)}+e^{(\lambda)}r\right)} + \frac{\sqrt{r}\frac{\partial}{\partial \rho}r}{4\left(ze^{(\lambda)}+e^{(\lambda)}r\right)}$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = & -\frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{r^4 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r \frac{\partial}{\partial \rho} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{3\,\rho r^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho r^3 \frac{\partial^2}{(\partial \rho)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z r^2 \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{r^3 \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{01} = & -\frac{i\,\rho z^2 r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{2i\,\rho z r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho r^3 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,r^3 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 r \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r^2 \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& + \frac{i\,r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r \frac{\partial}{\partial\rho} r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{02} = & \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^3 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial\rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial\rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial\rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial\rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{r^4 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial\rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{r^4 \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial\rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial\rho} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z r^2 \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{10} = {} & \frac{i\,\rho z^2 r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{2i\,\rho z r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,r^3 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho r \frac{\partial}{\partial\rho} r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{\rho z^2 r \frac{\partial}{\partial\rho}\widetilde{U}^2}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z r^2 \frac{\partial}{\partial\rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{\rho r^3 \frac{\partial}{\partial\rho}\widetilde{U}^2}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{\rho z^2 r \frac{\partial}{\partial z}\widetilde{U}^2}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z r^2 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& + \frac{\rho r^3 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{\rho z r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{\rho z r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{\rho r^2 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{z^2 r \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{z r^2 \frac{\partial}{\partial\rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{r^3 \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{\rho r \frac{\partial}{\partial\rho} r^2}{8\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z r \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho r^2 \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{3\,\rho r \frac{\partial}{\partial z} r^2}{8\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho z r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{\rho r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{z^2 \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{z r \frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{r^2 \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{3\,\rho r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{3\,\rho r}{8\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = {} & \frac{i\,\rho z^2 r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{2i\,\rho z r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho z r \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,r^3 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho r \frac{\partial}{\partial\rho} r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^3 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial\rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial\rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial\rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial\rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial\rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{r^4 \frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial\rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{r^4 \frac{\partial}{\partial\rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial\rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial\rho} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial^2}{(\partial\rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z r^2 \frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = & -\frac{i\,\rho z^2 r\,\frac{\partial}{\partial\rho}\widetilde{U}\,\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{2i\,\rho z r^2\,\frac{\partial}{\partial\rho}\widetilde{U}\,\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho r^3\,\frac{\partial}{\partial\rho}\widetilde{U}\,\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z r\,\frac{\partial}{\partial z}\widetilde{U}\,\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho r^2\,\frac{\partial}{\partial z}\widetilde{U}\,\frac{\partial}{\partial\rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r\,\frac{\partial}{\partial\rho}\widetilde{U}\,\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r^2\,\frac{\partial}{\partial\rho}\widetilde{U}\,\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho z r\,\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r^2\,\frac{\partial}{\partial\rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 r\,\frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z r^2\,\frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,r^3\,\frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z^2 r\,\frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r^2\,\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& + \frac{i\,r^3\,\frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r\,\frac{\partial}{\partial\rho} r\,\frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho r\,\frac{\partial}{\partial\rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2\,\frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z r\,\frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = & -\frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{r^4 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r \frac{\partial}{\partial \rho} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{3\,\rho r^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho r^3 \frac{\partial^2}{(\partial \rho)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r^2}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z r^2 \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{r^3 \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = & -\frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial \rho} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial z} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial \rho} r^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial \rho} r^2}{8\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial^2}{(\partial \rho)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r \frac{\partial}{\partial z} r^2}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r^2}{8\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^2 \frac{\partial}{\partial \rho} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{r^3 \frac{\partial}{\partial \rho} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2}{24\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & -\frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{2\,\rho z r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{3\,\rho r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{r^4 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z r^2 \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{r^3 \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_1 = {} & \frac{i\,\rho z^2 r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{2i\,\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho z^2 r \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& + \frac{2i\,\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z^2 r \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{2i\,\rho z r^2 \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,\rho r^3 \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho z r \frac{\partial}{\partial z}\lambda\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial z}\lambda\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial \rho}\lambda\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\lambda\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z^2 r \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial^2}{\partial \rho \partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho r^2 \frac{\partial^2}{\partial \rho \partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z \frac{\partial}{\partial \rho} r \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_2 = {} & \frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{4\,\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{5\,\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho r^3 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{2\,\rho z r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z r^3 \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{r^4 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial \rho} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{5\,\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho r^3 \frac{\partial^2}{(\partial \rho)^2} r}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{5\,\rho z r \frac{\partial}{\partial z} r^2}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{5\,z r^2 \frac{\partial}{\partial \rho} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{r^3 \frac{\partial}{\partial \rho} r}{6\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{5\,\rho r^2 \frac{\partial}{\partial z} r}{12\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^2}{3\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = {} & \frac{i\,\rho z^2 r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{2i\,\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho z^2 r \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& + \frac{2i\,\rho z r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} + \frac{i\,\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z^2 r \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{2i\,\rho z r^2 \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,\rho r^3 \frac{\partial^2}{\partial \rho \partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho z^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,\rho r^2 \frac{\partial}{\partial z}\widetilde{U}\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,\rho z r \frac{\partial}{\partial z}\lambda\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial z}\lambda\frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} \\
& - \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\widetilde{U}\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial \rho}\lambda\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\lambda\frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z^2 r \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho z r \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,z^2 r \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r^2 \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2} - \frac{i\,r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z r \frac{\partial^2}{\partial \rho \partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& - \frac{i\,\rho r^2 \frac{\partial^2}{\partial \rho \partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z \frac{\partial}{\partial \rho} r \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,\rho z \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,\rho r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} \\
& + \frac{i\,z^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} + \frac{i\,z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,z r}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)} - \frac{i\,r^2}{4\left(\rho z^2 e^{(2\,\lambda)} + 2\,\rho z e^{(2\,\lambda)} r + \rho e^{(2\,\lambda)} r^2\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = & -\frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U}^2}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{2\,\rho z r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& - \frac{\rho r^4 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho z^2 r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{2\,\rho z r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial \rho)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{\rho r^4 \frac{\partial^2}{(\partial \rho)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 r^2 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^3 \frac{\partial^2}{(\partial z)^2}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{\rho r^4 \frac{\partial^2}{(\partial z)^2}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 r \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial \rho}\widetilde{U} \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r^2 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial}{\partial \rho}\lambda \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{2\,\rho z r^2 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{3\,\rho r^3 \frac{\partial}{\partial z}\widetilde{U} \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda \frac{\partial}{\partial z} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z^2 r^2 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,z r^3 \frac{\partial}{\partial \rho}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{3\,r^4 \frac{\partial}{\partial \rho}\widetilde{U}}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} - \frac{\rho r^3 \frac{\partial}{\partial z}\widetilde{U}}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} + \frac{z^2 r^2 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{z r^3 \frac{\partial}{\partial \rho}\lambda}{\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3} \\
& + \frac{r^4 \frac{\partial}{\partial \rho}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r^2 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho r^3 \frac{\partial}{\partial z}\lambda}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{\rho z^2 \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{3\,\rho z r \frac{\partial}{\partial \rho} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z^2 r \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z r^2 \frac{\partial^2}{(\partial \rho)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z^2 \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{\rho z r \frac{\partial}{\partial z} r^2}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& - \frac{\rho z^2 r \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{3\,\rho z r^2 \frac{\partial^2}{(\partial z)^2} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^3 \frac{\partial^2}{(\partial z)^2} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{z^2 r \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} \\
& + \frac{z r^2 \frac{\partial}{\partial \rho} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} + \frac{r^3 \frac{\partial}{\partial \rho} r}{2\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho z r \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)} - \frac{\rho r^2 \frac{\partial}{\partial z} r}{4\left(\rho z^2 e^{(2\,\lambda)} r + 2\,\rho z e^{(2\,\lambda)} r^2 + \rho e^{(2\,\lambda)} r^3\right)}
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"